MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**C**
Course Name	**APPLIED SOIL ENGINEERING**	**Course Code**	**CIE 3122**
Semester	**V**	**Max Marks**	**50**
Date	**23-11-2024**	**Duration**	**3 HRS**

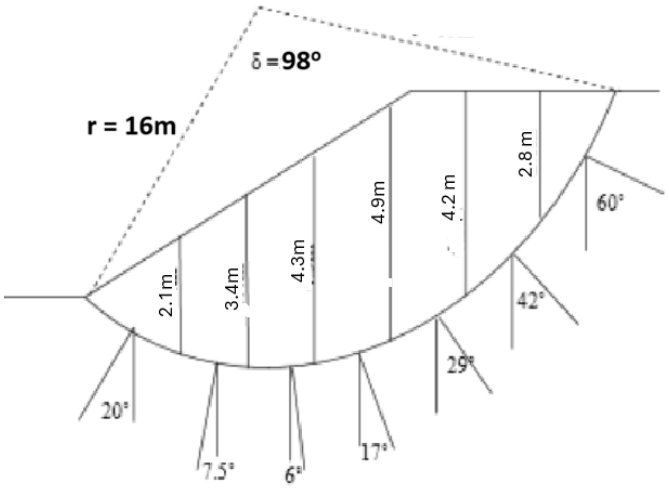
Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	Determine area ratio, inside clearance and outside clearance of the samplers given below. Comment on the sample disturbance from these values. All dimensions are in mm.	**5**	1	1,2	**3**

SamplerType	Driving Shoe		Sample tube	
	Outer diameter	Inner diameter	Outer diameter	Inner diameter
1	54	48	51	49
2	98	90	95	92
3	101	97	100	99

1B	Design a square footing to carry a safe load of 1400kN, on a sandy soil at a depth of 1.7m below ground level given unit weight of soil 17kN/m^3, standard penetration number = 23 and allowable settlement of 25mm. Water table is at base of footing and use FOS as 2.	**3**	4	1,2 ,3	**3**
1C	Discuss the causes and effect of negative skin friction on the pile load carrying capacity.	**2**	5	2,3	**3**
2A	A retaining wall of 7m high has two layers of backfill. The properties of soil are as follows. Top layer: 0 – 3m, γ= 18 kN/m^3, ϕ = 10º, c = 10kN/m^2. Bottom layer: 3 – 7m, γ= 19 kN/m^3, ϕ =20º, c = 5 kN/m^2. The water table is 4m below the top of the retaining wall. A uniform	**5**	2	1,2	**4**

	surcharge of magnitude 20 kN/m^2 acts over the backfill. Evaluate the total active earth pressure and its point of application after occurrence of tension cracks.				
2B	A retaining wall 6m high with horizontal backfill carries a uniform surcharge of 45kN/m^2. The backfill in top 3m is soil having $\phi = 36°$ and $\gamma = 19.8$ kN/m^3 and bottom 3m is soil with $\phi = 32°$ and $\gamma = 19$ kN/m^3. Illustrate the passive earth pressure distribution.	**3**	2	1,2	**4**
2C	The active earth pressure is decreased while the passive earth pressure is increased due to cohesion in soil. Justify.	**2**	2	1,2	**3**
3A	Determine the factor of safety against sliding for the slip surface shown in figure. The properties of soil are c= 22 kN/m^2, $\varphi = 25^\circ$ and $\gamma = 18.6$ kN/m^3. Use Swedish circle method of slices. The width of slice adjacent to toe is 3m and remaining slices are 4m wide. Provide a FBD for anyone slice. The mid-ordinate of slices shall be taken as the average of its sides. 	**5**	3	1,2,3,6	**4**
3B	Explain any three types of failures of finite slopes indicating the situations in which they are likely to occur.	**3**	3	1,6	**2**
3C	An highway embankment 8m high is proposed to be constructed with the soil properties c = 21 kN/m^2, $\phi = 10°$ and $\gamma = 17$ kN/m^3. Evaluate the inclination that can be provided if factor of safety with respect to cohesion is 1.5 and factor of safety with respect to friction angle is fully mobilised.	**2**	3	1,2,3,6	**4**
4A	A rectangular footing 1.5m × 3m is constructed at a depth of 1.5m in a soil deposit having c = 18kN/m^2, γ_{sat} = 20 kN/m^3, γ = 18kN/m^3 and ϕ = 25°. The ground water table is at foundation level. Evaluate the net	**4**	4	1,2,3,6,8	**3**

	ultimate bearing capacity by IS code method assuming local shear failure, taking factor of safety as 3.				
4B	Evaluate the safe load that a circular footing of diameter 2.2m can carry at a depth of 1.5m below ground level in a deposit of sand having γ_{sat} = 20 kN/m^3, γ = 17.5kN/m^3 and ϕ = 38˚. Water table is at 2m below ground level. Take FOS = 2.5. Use Terzaghi's method.	**4**	4	1,2,3,6	**3**
4C	Differentiate between uniform and differential settlements.	**2**	4	1,6	**2**
5A	A group of 16 piles of 35cm diameter, 9 m length and 75cm c/c spacing is subjected to a net load of 2500kN in a soil deposit having c = 80 kN/m^2, γ_{sat} = 20 kN/m^3, γ = 18kN/m^3, e_0 = 0.8 and Cc = 0.15. The hard strata is at 14m below ground level and ground water table is at 4m below ground level. Assume the load to be transferred at 2/3rd the length of the pile. Evaluate the consolidation settlement. Take two layers of equal thickness for settlement calculation.	**4**	5	1,2,3,6,8	**4**
5B	A group of 25 piles (square pattern) was installed in a layered clay deposit. The diameter of each pile is 450mm and c/c spacing is 1000mm. The length of the piles is 15m. The top layer is 6m thick with c_u=40kPa, m/α=0.9, ϕ=0 and the second layer is 14m thick with c_u=70kPa, m/α=0.8, ϕ=0. Estimate the safe load carrying capacity of the group with a factor of safety of 2.5. Assume Nc = 9.	**4**	5	1,2,3,6,8	**3**
5C	Under what conditions a pile foundation is more suitable than a shallow foundation?	**2**	5	1,6	**2**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***5***
2	***10***
3	***10***
4	***13***
5	***12***

BL	Total Marks
2	**9**
3	**22**
4	**17**
5	